

University of British Columbia LAND 304 – Winter 2026 || Page 1

LAND 304 - URBAN ECOSYSTEM RESILIENCE AND CLIMATE ADAPTATION 3 credits

Instructor:

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Acknowledgements:

As we gather for a course focused on urban ecosystem resilience and sustainability, it is important to reflect on the histories and cultures of the Indigenous communities that were present on the land before us. We recognize and honour the enduring presence and stewardship traditions of Indigenous communities across Canada. We acknowledge UBC's Point Grey Campus is located on the traditional, ancestral, and unceded territory of the xwm̓θkw̓y̓m (Musqueam) people. The forests, waterways, and coastal ecosystems that surround this region have long supported learning, stewardship, and community relationships with the environment.

Course description:

Urban ecosystems can provide important solutions to climate adaptation, biodiversity conservation, and human wellbeing through ecological planning, restoration, and stewardship. The goal of LAND 304 is to familiarize students with the fundamentals of ecosystem resilience, urban biodiversity, environmental monitoring, ecological restoration, and sustainability planning. The course explores interactions between environmental systems and urban development while considering ecological, social, and planning perspectives. Topics include environmental change, green infrastructure, ecosystem restoration, environmental governance, climate adaptation, and nature-based solutions. Special lectures will also explore Indigenous perspectives on stewardship and environmental decision-making. The course culminates with discussions of future cities, regenerative planning, and ecosystem-based approaches to climate resilience.

Course organization and Canvas website

- Lectures are on Tuesdays and Thursdays 1:00-2:20 PST online.
- Lectures, handouts and additional readings will be available on the UBC Canvas website.
- Lectures will be recorded and can be observed asynchronously.
- Students should check announcements on Canvas regularly throughout the term.
- Students may switch between live and recorded lectures throughout the semester.

Participation

- To earn participation marks, students must either attend the live lecture or watch the full lecture recording.
- Polling questions will be presented during lectures and within lecture recordings.
- Students watching recordings must complete lecture activities within two days to receive participation credit.
- Students are responsible for ensuring participation submissions are properly recorded.

Urban Ecosystem Reflection Assignment:

- Students will observe and document environmental features or sustainability initiatives within urban spaces.
- Reflections should connect observations to ecosystem resilience, biodiversity, and climate adaptation.
- Students may use environmental mapping or identification apps to support observations.
- Students are encouraged to discuss questions related to ecosystem identification with teaching assistants.

Academic objectives:

- (1) Understand urban ecosystems as interconnected adaptive systems;
- (2) Understand environmental resilience at local to regional scales;
- (3) Understand biodiversity, restoration, and ecosystem services in urban environments;
- (4) Understand ecological restoration and climate adaptation approaches;
- (5) Gain perspectives from Indigenous and community-based stewardship approaches;
- (6) Appreciate the importance of environmental planning and ecosystem-based management.

Recommended literature sources:

- Foster, H. & Kim, R. 2022. *Urban Ecology and Environmental Resilience*. Pacific Academic Press.
- Martin, S. 2020. *Climate Adaptation in Cities*. Greenleaf Publishing.
- Additional readings and case studies will be provided through Canvas.

Examinations, assignments and grading:

Participation 10%
Polling questions 10%
Urban ecosystem reflection assignment 15%
Midterm exam 25%
Final lecture exam 40%

**** NOTE,** The final exam is NOT cumulative.

**** NOTE,** Both the midterm and final exams MUST be taken to pass the course.

LAND 304: Tentative Lecture Schedule

Date Topic

Jan. 6 Introduction to urban ecosystems

Jan. 8 Climate adaptation and sustainability planning

Jan. 13 Urban biodiversity and ecological connectivity

Jan. 15 Green infrastructure systems

Jan. 20 Indigenous stewardship perspectives (Dr. Naomi Peters)

Jan. 22 Environmental monitoring and resilience indicators

Jan. 27 Urban water systems and watershed management

Jan. 29 Environmental governance and policy

Feb. 3 Ecological restoration strategies

Feb. 5 Midterm

Feb. 17 Community resilience and adaptation

Feb. 19 Environmental communication and engagement

Feb. 24 Urban forestry and climate mitigation

Feb. 26 Nature-based climate solutions

Mar. 3 Sustainable transportation systems

Mar. 5 Global environmental change and adaptation

Mar. 10 Environmental justice and equity

Mar. 12 Urban sustainability planning

Mar. 17 Case studies in ecosystem resilience

Mar. 19 Future cities and environmental innovation

Apr. 2 Study guide for final exam (Jordan Kim)

(FINAL EXAM TBA)

Communication protocols

Once the course begins, emails to instructors should be reserved for questions specific to your personal situation. Questions related to course content or deadlines are encouraged during lectures or through Canvas discussion spaces.

Academic and learning resources

UBC provides academic support, wellbeing resources, and accessibility services to support students throughout the term. Students experiencing difficulties are encouraged to communicate with the instructional team early.

Netiquette expectations

Students are expected to communicate respectfully and professionally in all online interactions and collaborative course activities.

Universities policies

UBC values academic honesty, accessibility, diversity, and respect for all members of the academic community.

Academic integrity

All students are expected to submit original work and appropriately acknowledge external ideas and sources. Academic misconduct may result in academic penalties.